Total Points: [10] [U1HW8 - W8 - Building a Bridge 1]

1. When you squeeze both ends of a popsicle stick together, what type of force is pressing the wood together?

- Deck
- Truss
- Span
- Buckling
- Joint
- Tension
- Compression

Answer:

2. Which of the following best defines the "Load" on a bridge?

- A pulling or stretching force that pulls a stick apart.
- When a stick bends sideways and snaps under a heavy push.
- The open distance between two tables or supports.
- The flat floor or roadway of the bridge.
- A strong bridge frame made of connected triangles.
- The place where two or more sticks are glued together.
- The heavy weight placed on top of the bridge.

Answer:

3. In a model popsicle stick bridge, what is a "Joint"?

- The flat floor or roadway of the bridge.
- The place where two or more sticks are glued together.
- When a stick bends sideways and snaps under a heavy push.
- The heavy weight placed on top of the bridge.
- How strong your bridge is compared to its weight.
- A pushing or squeezing force that presses a stick together.
- The open distance between two tables or supports.

Answer:

4. What is the "Deck" of a bridge?

- The place where two or more sticks are glued together.
- The flat floor or roadway of the bridge.

• A pulling or stretching force that pulls a stick apart.

• The open distance between two tables or supports.

• The heavy weight placed on top of the bridge.

• A strong bridge frame made of connected triangles.

• When a stick bends sideways and snaps under a heavy push.

Answer:

5. When a heavy load is placed on the center of a bridge, how do the forces act on the top and bottom sections?

• The top sticks buckle sideways because of strong tension forces.

• The bridge experiences no pushing or pulling forces until it breaks.

• Both the top and bottom sticks are pulled apart by tension.

• Both the top and bottom sticks experience only buckling and no forces.

• The top is pulled apart by tension, and the bottom is pushed together by compression.

• The top of the bridge is pushed together (compression), and the bottom is pulled apart (tension).

• Both the top and bottom sticks are pushed together by compression.

Answer:

6. What is the "Efficiency Score" and how is it calculated for a model bridge?

• The number of triangles multiplied by the number of glue joints.

• How strong your bridge is compared to its weight (Weight Held / Bridge Weight).

• The total time in minutes the bridge holds weight before breaking.

• The bridge weight divided by the total weight held.

• The weight of the bridge minus the total load held.

• The total length of the span divided by the bridge width.

• The total number of sticks used multiplied by the span length.

Answer:

7. What is a "Truss" in bridge design?

• When a stick bends sideways and snaps under a heavy push.

• The place where two or more sticks are glued together.

• The flat floor or roadway of the bridge.

• The open distance between two tables or supports.

• A strong bridge frame made of connected triangles.

• A pulling or stretching force that pulls a stick apart.

• The heavy weight placed on top of the bridge.

Answer:

8. Which statement accurately defines "Tension" in a bridge structure?

• The flat floor or roadway of the bridge.

• A pulling or stretching force that pulls a stick apart.

• When a stick bends sideways and snaps under a heavy push.

• The open distance between two tables or supports.

• How strong your bridge is compared to its weight.

• A pushing or squeezing force that presses a stick together.

• A strong bridge frame made of connected triangles.

Answer:

9. What does the term "Buckling" mean when testing a bridge member under a heavy load?

• The open distance between two tables or supports.

• A strong bridge frame made of connected triangles.

• The place where two or more sticks are glued together.

• A pulling or stretching force that pulls a stick apart.

• How strong your bridge is compared to its weight.

• When a stick bends sideways and snaps under a heavy push.

• The flat floor or roadway of the bridge.

Answer:

10. What does the "Span" of a bridge refer to?

Total Points: [10] [U1HW8 - W8 - Building a Bridge 1]

• The place where two or more sticks are glued together.

• A strong bridge frame made of connected triangles.

• The flat floor or roadway of the bridge.

• The heavy weight placed on top of the bridge.

• The open distance between two tables or supports.

• A pushing or squeezing force that presses a stick together.

• How strong your bridge is compared to its weight.

Answer:

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